Effects of Social Separation on Immune Function and Brain Neurotransmitters in Cebus Monkey (*C. Apella*)[a]

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INTRODUCTION

This study examined the influence of 3–5 days of social separation upon measures of immune function and central nervous system neurotransmitters in new world primates (*C. apella*). Immune function was assessed by measuring total lymphocyte counts and both the percentages and absolute values of lymphocyte subsets (T helpers, T suppressors, natural killer cells, and B cells). These measurements were accomplished using flow cytometry and human T cell markers (CD-4, CD-8, Leu-11, and HLA-DR). In addition, functional assessments of the immune system were made by measuring phytohemagglutinin (PHA)-induced blastogenesis. Finally, the immune system was challenged with the benign antigen, keyhole limpet haemocyanin (KLH, 3.5 mg/kg), and the above measures taken serially.

Concurrent extracellular measures of brain neurotransmitters, and their metabolites, were undertaken employing microdialysis of the amygdala and hypothalamus. Cortisol levels in fecal excretion were measured to assess levels of "stress" during social separation (SS) and group housing (G).

METHODS

Feral born *C. apella* (4 male and 4 female, 2–5 kg) were group-housed in a large outdoor enclosure where they were adapted for 3–12 months prior to examination. The animals were maintained on a diet of new world monkey chow (Purina) and fresh fruit under all conditions. 3–5 ml of blood was withdrawn from the saphenous vein whenever the immune system was assessed, and fecal samples were frozen

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for subsequent standard radioimmunoassay (Pantex, Santa Monica, CA). Two subjects were implanted with chronic guide cannula over their amygdala, while two other subjects had cannulae implanted over both the amygdala and hypothalamus. When not being dialyzed, the guide cannulae were protected by a short length of surrounding plastic PVC pipe and threaded cap. During dialysis (Carnegie Medicin, Solna, Sweden; Bioanalytical Systems, Inc., West Lafayette, IN), the animals were comfortably seated in a primate restraining chair, and the cannulae obdurators replaced by dialysis probes. When subjects were dialyzed following SS, the animals were isolated by a screen surrounding their chair. Otherwise, the animal was dialyzed while facing a chaired conspecific from the group. Serial fractions of perfusate (50 μl) were collected over a four-hour period at a flow rate of 3 μl/min. 25-μl aliquots of perfusate were analyzed for biogenic amines, and their metabolites, via HPLC (Spectra Physics), employing a BAS electrochemical detector.

Data for this experiment was obtained over a period of a year and a half. After two baseline measures of the immune system, two male and two female subjects underwent SS at 21, 85, and 251 days from the start of the experiment. Throughout this interval eight additional samples were drawn at various intervals during group housing.

RESULTS

SS resulted in a repeatable and reversible drop in total lymphocyte count (FIG. 1a) and selective increase in the percentage, but not total number, of T suppressor markers. These changes slowly returned to baseline values following return to the group. During SS, NK cell markers declined in number, but their percentage of total lymphocyte counts were variable. The response of the T helper cells was also variable, while the B cell markers showed no response. PHA-stimulated blastogenesis was reduced in these and all subsequent experimental animals undergoing SS (FIG. 1c). Following SS, fecal cortisol levels were elevated, with females tending to show the greatest response. This difference in cortisol response was reflected in the elevation in T suppressor markers obtained during SS.

SS resulted in an increase in NE, 5-HIAA, and HVA in the amygdala, while DOPAC decreased. NE and HVA also increased in the hypothalamus, while a decrease in 5-HIAA was opposite to that observed in the amygdala (FIG. 2a,b).

Inoculation with KLH produced a robust antibody response in all subjects, with titers increasing from 1/32 to between 1/512 and 1/4096. Following SS, four subjects were inoculated with KLH, and remained separated for an additional three days. Two additional subjects were inoculated while remaining in the group cage. These latter subjects showed a 600 percent increase in PHA-stimulated blastogenesis, while the separated subjects showed no change in blastogenesis (FIG. 1d). Similarly, T helper markers increased dramatically in those animals inoculated in the group setting, while no response was observed in animals undergoing SS. In neither condition were T suppressor markers affected by KLH. KLH produced a decrease in 5-HIAA in the amygdala, and reversed the direction of effect of SS on both 5-HIAA and HVA, with large reductions in both metabolites being observed. In the hypothalamus, KLH inoculation appears to have potentiated the drop in 5-HIAA produced by SS. HVA undergoes a dramatic elevation with KLH inoculation. However, the effects of SS are reversed with KLH (FIG. 1b). SS alone appears to elevate HVA. If KLH is given during this period, however,

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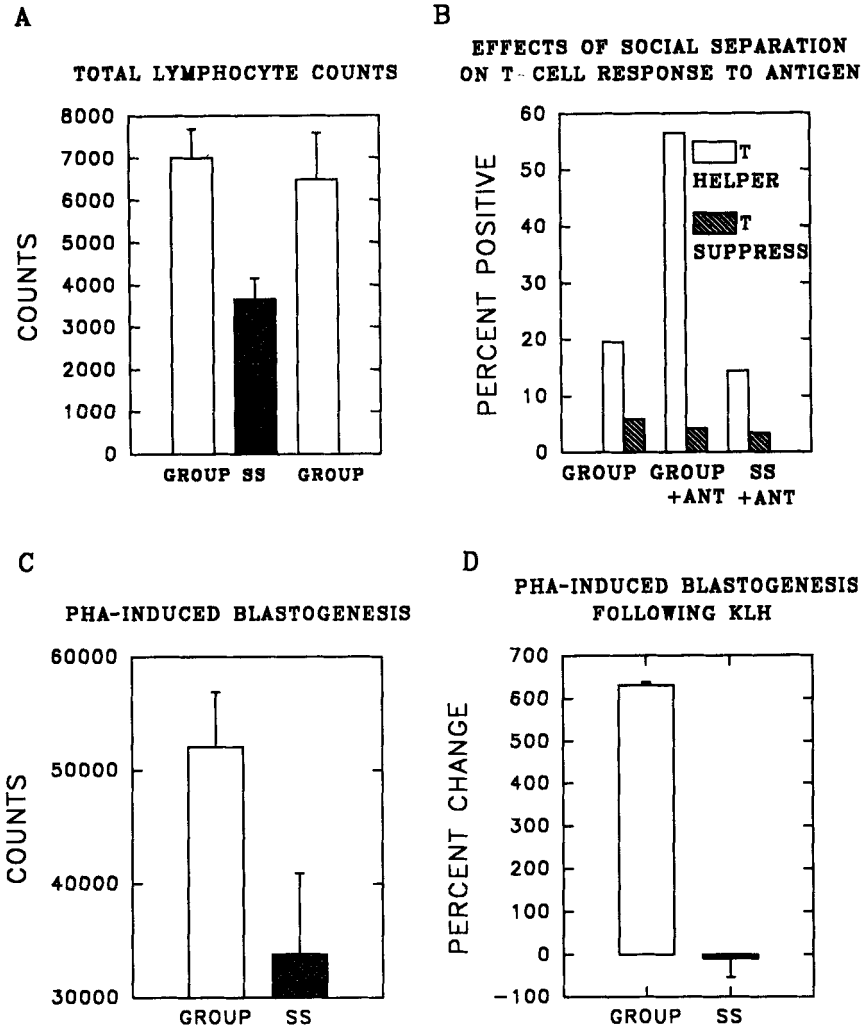


FIGURE 1. **(A)** Total lymphocyte counts of all animals taken before SS, at the end of SS, and following recovery. **(B)** Percent of lymphocytes with markers for T helper or T suppressor cells taken from animals while in the group setting before and after inoculation with antien, and after inoculation with antigen while in SS. **(C)** PHA-induced blastogenesis in all animals while in the group or following SS. **(D)** PHA-induced blastogenesis following KLH inoculation in the group or following SS.

the KLH-induced elevation in HVA is not observed, and a lower amount of HVA is recovered than in any other condition.

In the amygdala, both treatments produce an increase in NE and a decrease in DOPAC. In addition, there appears to be a summation of effects for both mono-

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amines. These effects are partially reflected in the dialysate from the hypothalamus. On the other hand, the 5-HIAA and HVA responses in these two structures demonstrated a great deal of independence from one another (FIG. 2a,b).

DISCUSSION

In human studies, chronic stress,[1] depression,[2] and bereavement[3] all result in a decrease of PHA-stimulated blastogenesis, consistent with the results of SS in our monkeys. In addition, SS was also found to block the increase in blastogenesis produced by inoculation with KLH. As with human depression,[4] SS resulted in a lymphopenia. However, the increase in T suppressor cell markers and the increase

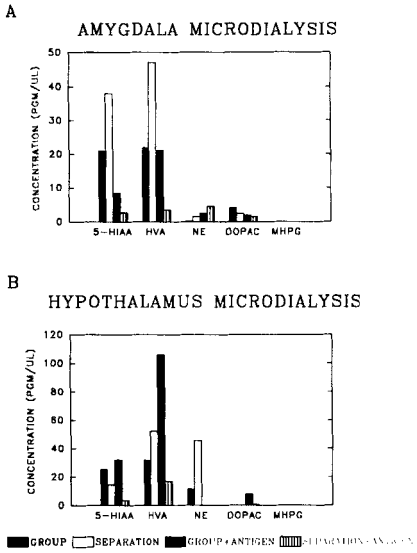


FIGURE 2. Dialysis of **(A)** amygdala and **(B)** hypothalamus while residing in group and following SS, with and without antecedent KLH inoculation.

in the suppressor/helper ratio which we have observed has not previously been reported. Chronic stress in humans[1] has also been associated with absolute decreases in NK, B, and Pan-T cell levels. The changes observed in the dialysate from the amygdala and (to a lesser extent) the hypothalamus, following SS, have been observed in the CSF of depressed patients.[5] Most striking, perhaps, are the interactions which we have observed between SS and KLH inoculation in these measures. We have shown concomitant changes in immune measures and CNS monoamines during SS and following KLH inoculation. Furthermore, we have demonstrated interactions between social "stress" and immune challenge in both these domains. Taken together, these effects suggest that the immune and central nervous systems exert mutual influences upon one another and that emotional stress in the central domain or challenge in the immune domain can influence

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functioning in the other domain. The significance of these interactions for behavior remains to be elaborated.

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